

Digital Diary

PYTHON PROJECT | SESSION 1

Building Your Own Contact Book [cite: 287, 288]

The Lost Number Problem

Imagine you forgot your friend **Aarav's** phone number and need to call him for homework help! [cite: 290] How do we keep all important numbers safe and easy to find? [cite: 291]

WHAT IS A DIGITAL DIARY?

A Digital Diary is like a modern address book on your computer! [cite: 293] It uses a special Python tool called a **Dictionary** to store information as **Keys** and **Values**. [cite: 293, 307]

Keys and Values

Think of it this way: **Key = Label**, and **Value = Information Stored!** [cite: 296]

INDIAN EXAMPLES [CITE: 294, 295]

- **Kirana store ledger:** Customer name (Key) → Amount owed (Value)
- **Aadhaar card:** Unique number (Key) → Your details (Value)
- **Mobile contacts:** 'Mummy' (Key) → Phone number (Value)

PYTHON DICTIONARIES REFRESHER

```
contacts = {  
    'Daddy': '9876543210',  
    'Mummy': '9123456789',  
    'Diya': '9988776655'  
}
```

You already know how to use these! [cite: 309]

Let's Practice!

Here are some contacts: [cite: 311]

- Aarav: 9876543210 [cite: 312]
- Saanvi: 9123456789 [cite: 313]
- Arjun: 9988776655 [cite: 314]

Question: Can you tell the phone number for Saanvi? [cite: 315]

Starting Our Project

Let's begin with an empty dictionary: [cite: 318]

```
contacts = {}
```

OUR MENU OPTIONS [CITE: 319]:

1. Add Contact [cite: 320]
2. View All Contacts [cite: 321]
3. Search Contact [cite: 322]
4. Exit [cite: 323]

Adding Contacts

Here is how we add a new contact to our dictionary step by step! [cite: 324, 326]

```
def add_contact():  
    name = input("Enter name: ")  
    phone = input("Enter phone number: ")  
    contacts[name] = phone  
    print("Shabaash! Contact added.")
```

The Science: The **name** becomes the key, and the **phone number** becomes the value!
[cite: 331, 332]

Viewing All Contacts

The 'for loop' goes through each contact one by one and prints the name with its phone number! [cite: 337]

```
def view_contacts():  
    for name, phone in contacts.items():  
        print(f"{name}: {phone}")
```

Searching Contacts

The Truecaller Logic: Just like Truecaller helps you find who's calling, our search feature finds contacts instantly! [cite: 346, 347]

```
def search_contact():  
    name = input("Enter name: ")  
    if name in contacts:  
        print(f"Number: {contacts[name]}")  
    else:  
        print("Contact not found.")
```

Our Menu Loop

Think of a **while loop** as a door that stays open until you decide to leave! [cite: 340] Our menu keeps showing options until you choose Exit. [cite: 341]

```
while True:
    print("1. Add 2. View 3. Search 4. Exit")
    choice = input("Pick: ")
    if choice == '4':
        break
```

RECAP: WHAT WE BUILT [CITE: 354]

- Used dictionaries to store contacts.
- Keys = Names, Values = Phone numbers.
- Built Add, View, and Search functions.
- Created a UI loop to keep the app running.

Shabaash!

You built your first digital diary! [cite: 355]

NOW IT'S YOUR TURN! [CITE: 356]

Start coding your contact book and make it yours. [cite: 356] Be creative, explore new ideas, and don't be afraid to try different things. [cite: 357]

You have all the skills you need! [cite: 357]

Thank you for coding with us today!

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